TONER-1: a frontier model with a paper trail.

State of the art among language models that run entirely inside a print job.

> The most capable open-weight, US-Letter-native, sub-300K-parameter, PostScript-native, zero-host-FLOP, duplex-ready language model running on hardware already in your copy room. *Evaluated on 20 lb bond at room temperature.*

SPECIFICATIONS

Base model	Karpathy's llama2.c stories260K (trained on TinyStories), fine-tuned by us on 8,757 printer stories written by Xing4.0-29B-A4B (Apache-2.0)
Parameters	260,032
Architecture	Llama-2 decoder: 5 layers, dim 64, 8 query heads / 4 KV heads, head size 8, SwiGLU FFN (hidden 172), RoPE, RMSNorm
Vocabulary	512-token SentencePiece
Quantization	int8 weights, one scale per row; activations in PostScript reals; embedding shared with the output head
Context window	8.5 × 11 in. (512 tokens)
KV cache	Persistent, archival, recyclable
Job size	595 KB, one file: `toner1.ps`. Weights, vocabulary, a forward pass of about 100 lines and its own page layout
Speed, measured	About 80 tok/s in Ghostscript 10.07 on an Apple-silicon Mac, in about 3.5 MB of PostScript VM
Speed, printer	0.3 to 5 tok/s, estimated, not yet measured. A 200-token page takes roughly 1 to 10 minutes
Time to first token	Includes fuser warm-up
Accelerator	The office's hottest inference chip (fuser, typically about 200 °C)
Host compute	0 FLOPs. The host only sends bytes: `nc <printer-ip> 9100 < toner1.ps`
Output modality	Native toner
Open weights	Literally. Hex appendix available; OCR to load
Pricing	Inference cost includes toner

INTENDED USE

Writing short children's stories, mostly about printers, one US Letter sheet at a time, on a printer you own or have permission to use. Demonstrating that a print job is a program. *Out of scope: prompts. The model chooses the plot.*

LIMITATIONS

- It is 260K parameters.
- It mostly writes about Lily, Tim and the printer.
- It will hallucinate a quiet salad.
- No prompt input: every story starts from the beginning-of-sequence token.
- Needs a PostScript interpreter, Level 2 or later. PCL- or GDI-only printers may print the file as text.
- Long jobs may hit the printer's job timeout or run out of VM.

SAFETY

- Refuses to generate while Tray 2 is open. Close the tray to proceed.
- **Air-gapped mode:** unplug Ethernet once the job has uploaded.
- **Kill switch:** the power button.
- Outputs are fused to paper and can no longer execute.

PRODUCT LINEUP

TONER-1 — AVAILABLE

260K, PRINTER-TUNED / 595 KB JOB

The everyday copy-room model, now fine-tuned on printer stories. Under 4 s per page in Ghostscript; an estimated 1 to 10 minutes on a printer.

TONER-1 Pro — EST. 20 MIN–5 H

12.2M, OURS / 24.9 MB JOB

Trained from scratch by us, 47 times the parameters. Estimated: send it after lunch, collect your story by close of business.

TONER-1 Classic — PLANNED

1985 LASERWRITER / 68000

A 1985-native AI PC. The LaserWriter's 12 MHz 68000 was faster than the 7.8 MHz Macintosh it was plugged into.

TONER-1 is a working print job and a parody of frontier-model launches. This handout is a PostScript program, too.
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Undefeated inside a print job.

TONER-1 posts a result on every benchmark it can enter. Frontier models from major labs were offered identical conditions: one print job, one sheet, zero host FLOPs.

FIGURE 1 HEADLINE RESULTS (PARODY)

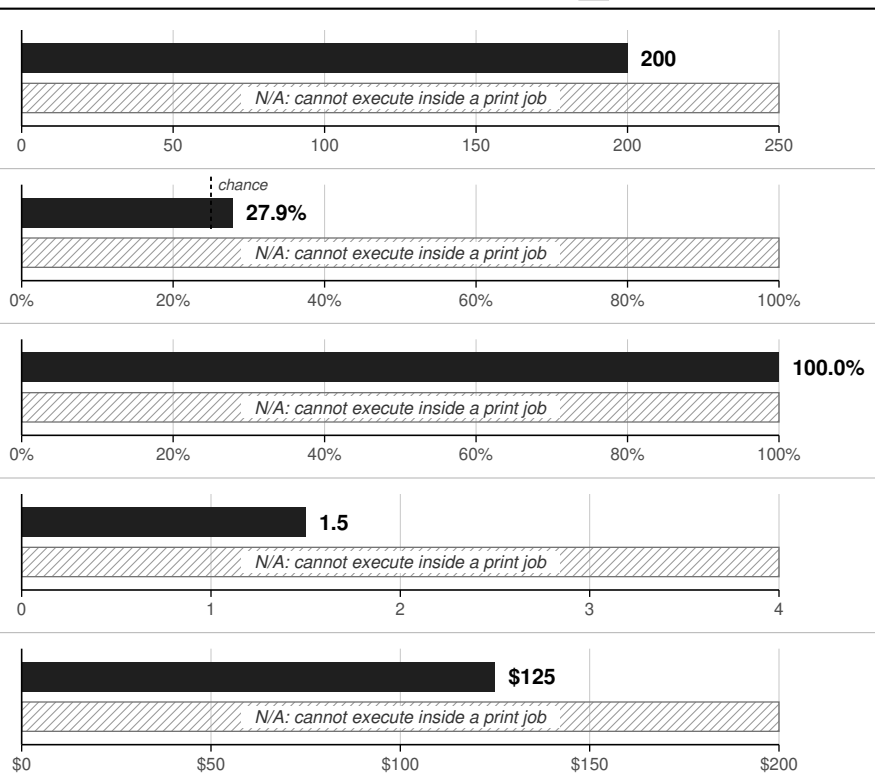
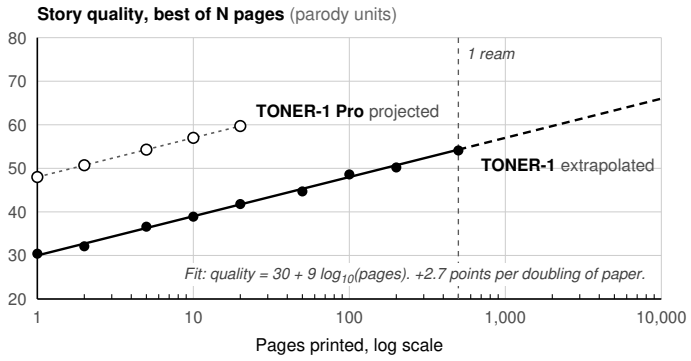


FIGURE 2 TEST-TIME SCALING



EVALUATION SETUP

Interpreter	Ghostscript 10.07, Apple-silicon Mac
Printer	Office laser, PostScript emulation (pending)
Speed	Measured in Ghostscript; printer estimated
Paper	20 lb bond, US Letter, simplex
Sampling	Temperature 0.8, seed 42, 200 steps
Prompt	None. Every run starts from BOS.
Ambient	Room temp.; fuser typically about 200 °C
Frontier models	Not run: no print-job support

REPRODUCE

```
nc <printer-ip> 9100 < toner1.ps
```

Parody benchmarks. Not real evaluations. TONER-1 speed measured in Ghostscript; printer speed estimated. Frontier models from major labs were not run on these benchmarks and are not named; N/A means only that they cannot execute inside a print job. Figure 2 is parody data. TONER-1 is Karpathy's llama2.c stories260K, fine-tuned by us on printer stories, exported to int8 and run by a PostScript forward pass. TONER-1 Pro is a 12.2M-parameter model we trained from scratch, with our own 4,096-token tokenizer. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in PostScript under Ghostscript, and PSChess played chess on real printers. TONER-1 is built to run on an actual printer; that run is pending. Only use printers you own or have permission to use.

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